

# XML

- XML stands for Extensible Markup Language. defines a set of rules <sup>for</sup> encoding document
- XML was designed to store & transport data. b/w two application <sup>→ that in both human & machine readable</sup>
- XML is a software & hardware independent tool for carrying information.
- XML tags are not predefined. You must define your own tags.
- XML is designed to be self-descriptive.
- XML does not do anything. It is just information wrapped in tags.

## XML document Example

<note >

<to> Pak Army </to>

<from> Imran Khan </from>

<heading> Reminder </heading>

<body> Release me! </body>

</note >

## Rules of XML syntax

1. Only one root tag.
2. Case sensitive
3. Every opening tags needs a closing tag.
4. Attribute values must be <sup>quoted</sup>



## Difference b/w XML & HTML

XML & HTML were designed with different goals:

- XML was designed to carry data - with focus on what data is
- HTML was designed to display data - with focus on how data looks
- XML tags are not predefined like HTML tags. User can define own tags.

### XML is Extensible

Most XML application will work as expected even if new data is added or removed.

For example

Imagine an application designed to display the original version of note.xml (`<to>`, `<from>`, `<heading>`, `<body>`)

Then imagine a new version of note.xml added `<date>` and `<hour>` element and removed `<heading>`.



The way XML is constructed, old version of the application can still work.

## XML Simplifies Things

- XML simplifies data sharing
- XML simplifies data transport
- XML simplifies platform changes
- XML simplifies data availability

Many computer systems contain data in incompatible formats. Exchanging data b/w incompatible system/upgraded systems is a time-consuming task for web developers. Large amount of data must be converted and incompatible data is often lost.

XML stores data in plain-text format. This provides a software & hardware independent way of storing, transporting and sharing data.

XML makes it easier to expand or upgrade to new OS, new applications or new browser without losing data.



With XML, data can be available to all kinds of "reading machines" like people, computers, voice machines, new feeds etc.

## How Can XML be used?

XML separates data from presentation

→ XML does not carry information about how to be displayed.

→ The same XML data can be used in many different presentation scenarios

Because of this, with XML, there is full separation b/w data & presentation

XML is often a complement to HTML

In many HTML application, XML is used to store or transport data, while HTML is used to format and display the same data

XML separates data from HTML

When displaying data in HTML, you should not have to edit



the HTML file when the data changes.  
With XML, the data can be stored in  
separate XML files.

With a few lines of JavaScript code,  
you can read an XML file and update  
the data content of any HTML page.

### Transaction Data

Thousands of XML formats exist, in many  
different industries to describe day-to-day  
data transactions:

- Stocks and shares
- Medical data
- Financial transactions
- News info.
- Mathematical data
- Weather services
- Scientific measurements

### XML Tree

XML document form a tree structure that  
starts at "root" and branches to "leaves".

→ First line in XML declaration defines its version

→ Next line describe root element and next

lines are child element. Parent, child,

siblings are used to describe relationship b/w  
element.



# XHTML

XHTML stands for Extensible Hyper Text Markup Language.  
→ XHTML derived from HTML to follow XML standards.

→ XHTML combine the flexibility of HTML with the stricter syntax rule of XML.

→ It follows XML rule, such as requiring all elements to be properly nested and closed, elements must be in lowercase, attribute names must be in lowercase and their value must be quoted, making it more consistent and easier to process by XML parsers.

- XHTML documents are written using XML syntax which makes them compatible with other XML-based technologies and tools.

- `<!DOCTYPE >` is mandatory

- The `xmlns` attribute in `<html>` is mandatory



- `<html>`, `<head>`, `<title>`, `<body>` elements are mandatory.

Application:

- Web development, especially for creating well-structured and semantic web pages.
- Mobile web development, as it facilitates the creation of web pages that are compatible with various devices and platforms.
- Integration with XML-based technologies and tools for processing and manipulating web content.



## XHTML MP (Mobile Profile)

XHTML MP is a subset of XHTML, which is stricter version of HTML.

XHTML MP is a mobile-specific version of XHTML designed for mobile devices with limited resources and capabilities.

→ It is designed for delivering web content to mobile devices.

→ XHTML ~~and~~ MP based on XHTML and optimized for mobile browsing, with features like small screen rendering, limited bandwidth, simplified navigation.

→ It is designed to provide a consistent user experience across different mobile devices and platforms.

### Application

It is used in various mobile web development frameworks and platforms to create mobile-friendly websites and web applications.



Rendering refers to process of converting HTML, CSS into visual layout & design that users see on their screen.

Mobile application services, providing a standard markup language for delivering content to mobile user.

Integration with mobile browsers and platforms, ensuring compatibility and consistency in rendering web content on various mobile devices.